

R - Programming

UNIT 5: Data Science Tools and Case Studies

MA10509: MATHEMATICS FOR DATA SCIENCE

Introduction to R

- **R** is a programming language and software environment used for **statistical computing and data analysis**.
- It is **open-source** and freely available.
- Developed by **Ross Ihaka** and **Robert Gentleman** at the University of Auckland, New Zealand.
- R is widely used by statisticians, data analysts, and researchers for data visualization, statistical modeling, and machine learning.

Features of R

- Open-source and free to use.
- Supports data handling and storage facilities.
- Includes a wide variety of statistical and graphical techniques.
- Extensible through packages.
- Provides powerful data visualization tools.
- Cross-platform (works on Windows, macOS, Linux).

Installing and Starting R

- **Download and install R:** from <https://cran.r-project.org>
- **RStudio** is a user-friendly IDE for R, available at <https://posit.co/download/rstudio/>
- After installation, open **RStudio** and start writing commands in the **Console**.

Basic Syntax and Operations

(a) Assignment Operator

$x \leftarrow 5$

$y = 10$

$z \leftarrow x + y$

$\text{print}(z)$

(b) Arithmetic Operations

Operator	Description	Example
+	Addition	4 + 5
-	Subtraction	10 - 2
*	Multiplication	3 * 4
/	Division	10 / 2
^	Exponentiation	2 ^ 3

(c) Comments

This is a single line comment

5. Data Types in R

Type

Numeric

Integer

Character

Logical

Complex

Example

2.5, 10

5L

"Hello"

TRUE, FALSE

2 + 3i

Example:

```
a <- 10
```

```
b <- "R Programming"
```

```
c <- TRUE
```

6. Data Structures in R

(a) Vector

One-dimensional array containing elements of the same type.

```
v <- c(1, 2, 3, 4, 5)  
print(v)
```

(b) Matrix

Two-dimensional data structure with rows and columns.

```
m <- matrix(1:9, nrow=3, ncol=3)  
print(m)
```


6. Data Structures in R (...cont)

(c) List

Collection of different types of elements.

```
l <- list(name="Shiva", age=20, marks=c(90, 85, 88))  
print(l)
```

(d) Data Frame

Table-like structure where each column can be of different type.

```
students <- data.frame(Name=c("Amit", "Neha", "Ravi"),  
                      Marks=c(85, 90, 78),  
                      Grade=c("A", "A+", "B"))  
print(students)
```

7. Control Structures

(a) If-Else Statement

```
x <- 10  
if(x > 0){  
  print("Positive number")  
} else {  
  print("Non-positive number")  
}
```

7. Control Structures (...cont)

(b) **For Loop**

```
for(i in 1:5){  
  print(i)  
}
```

(c) **While Loop**

```
count <- 1  
while(count <= 5){  
  print(count)  
  count <- count + 1  
}
```

8. Functions in R

Functions are used to perform specific tasks.

```
add <- function(a, b){  
  result <- a + b  
  return(result)  
}  
add(5, 3)
```

9. Reading and Writing Data

(a) Reading CSV File

```
data <- read.csv("data.csv")  
print(head(data))
```

(b) Writing CSV File

```
write.csv(data, "output.csv", row.names = FALSE)
```

10. Basic Statistical Functions

Function	Description	Example
<code>mean()</code>	Average	<code>mean(c(10, 20, 30))</code>
<code>median()</code>	Middle value	<code>median(c(1, 3, 5, 7, 9))</code>
<code>sd()</code>	Standard deviation	<code>sd(c(4, 8, 12))</code>
<code>var()</code>	Variance	<code>var(c(4, 8, 12))</code>
<code>sum()</code>	Sum of elements	<code>sum(c(2, 4, 6))</code>
<code>length()</code>	Number of elements	<code>length(c(2, 4, 6))</code>

11. Data Visualization

R provides various plotting functions.

(a) Basic Plot

```
x <- c(1,2,3,4,5)
```

```
y <- c(2,4,6,8,10)
```

```
plot(x, y, type="o", col="blue", main="Simple Line Plot")
```

11. Data Visualization (...cont)

(b) **Bar Plot**

```
barplot(c(10,20,30), names.arg=c("A","B","C"), col="green",  
main="Bar Plot")
```

(c) **Histogram**

```
hist(c(1,2,2,3,3,3,4,4,5), main="Histogram Example",  
col="orange")
```


12. Packages in R

- Packages extend R's functionality.
- To install and load packages:

install.packages("ggplot2")

library(ggplot2)

13. Applications of R

- Statistical analysis and hypothesis testing
- Data visualization and reporting
- Machine learning and predictive modeling
- Data cleaning and preprocessing
- Academic research and simulations

14. Summary

- R is a powerful language for data analysis and statistics.
- Basic understanding of syntax, data structures, and visualization helps beginners build analytical skills.
- RStudio makes coding and data handling user-friendly and efficient.

List of R-Programs

1. Display “Hello World”
2. Add two numbers
3. Find square and cube of a number
4. Find square and cube of a number
5. Find largest of three numbers
6. Calculate area of circle
7. Print numbers from 1 to 10
8. Sum of first N natural numbers
9. Generate multiplication table
10. Find factorial of a number

List of R-Programs (...cont)

11. Calculate Simple Interest
12. Check if a number is Prime
13. Create and display a vector
14. Find Maximum and Minimum of a vector
15. Display elements of a matrix
16. Calculate mean of numbers
17. Use a function to add two numbers
18. Plot a Simple Graph
19. Plot a Bar graph
20. Plot a Histogram